Data Visualization and Presentation

DATA SCIENCE BOOTCAMP

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Objectives

1. Understand the purpose of data visualization.
2. Distinguish between exploratory and explanatory visuals.
3. Select an appropriate visual for the data type and goal.
4. Learn techniques to highlight key insights.
5. Understand methods of presenting a convincing argument using data.

Data Visualization

Effective data visualization can help make insights accessible to your audience.

When presenting findings, your goal is to simplify complex information, highlight key takeaways, and support your main arguments.

When analyzing data, you have likely made a lot of exploratory plots. When presenting it, you should focus on choosing the few that you think most clearly convey your message and work to polish them to most effectively communicate your takeaways.

If you're including a plot, you should have a good reason for doing so.

Exploratory vs. Explanatory Visualization

Exploratory data visualization is used as a tool to assist in the analysis of data.

- Primarily done for the analyst or data scientist to understand and identify patterns, trends, or anomalies.
- Design is less important than quick iteration.
- Can use more dense visuals or more of the data in order to encourage discovery.

Explanatory data visualization is used to communicate an insight or pattern from the data.

- Created with a specific audience in mind, focusing on conveying a clear and convincing message.
- Good design is crucial.
- More polished, often using a more refined selection of data to focus on on specific elements to make insights easy to understand at a glance.

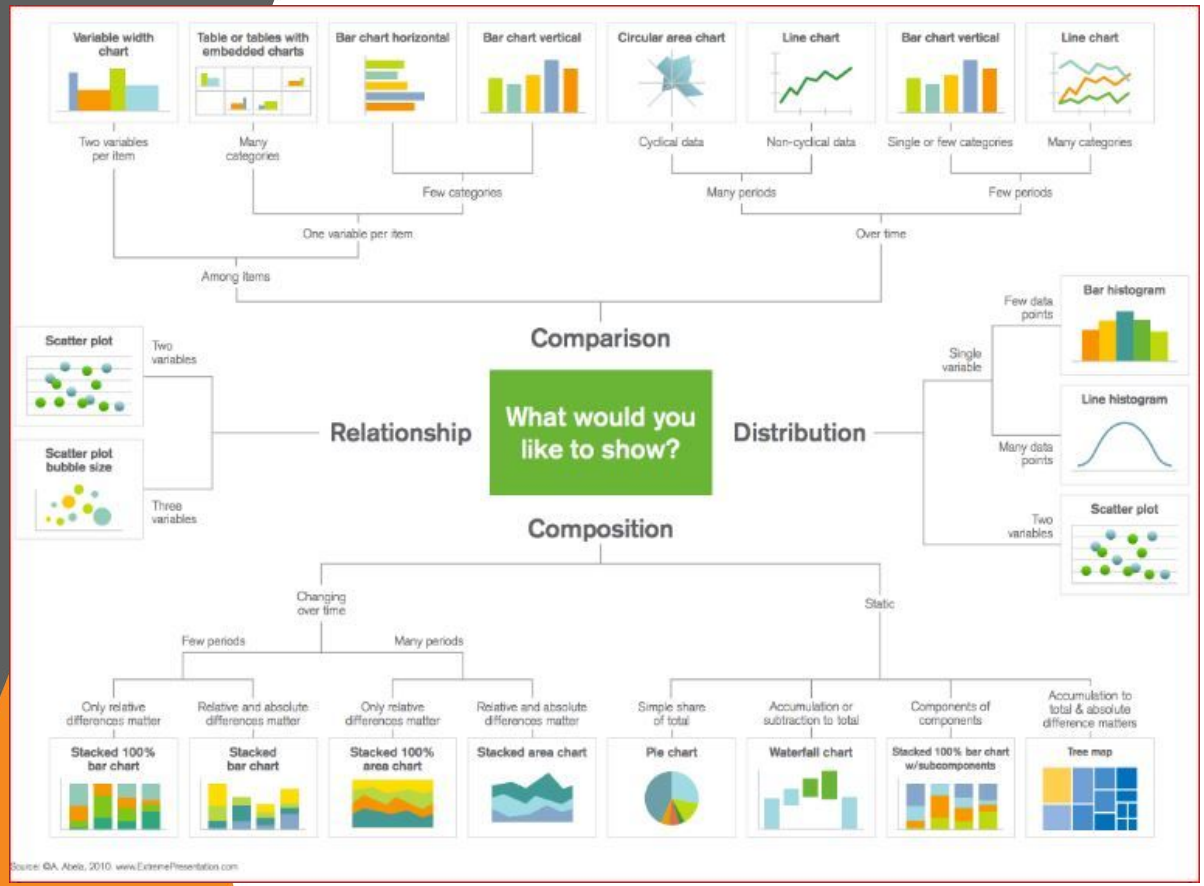
Choosing the Right Visualization

Choose the right type of plot for the type of data.

- Bar charts for categorical
- Box plots or histograms for numeric variables
- Scatterplots for comparing numeric variables
- Line charts for trends

Consider if you're wanting to convey a distribution, relationship, or comparison.

Avoid overly complex visuals, especially for a presentation format.



https://python-graph-gallery.com/

https://blogs.sap.com/2016/12/05/what-how-would-you-like-to-show/

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Highlighting Key Insights

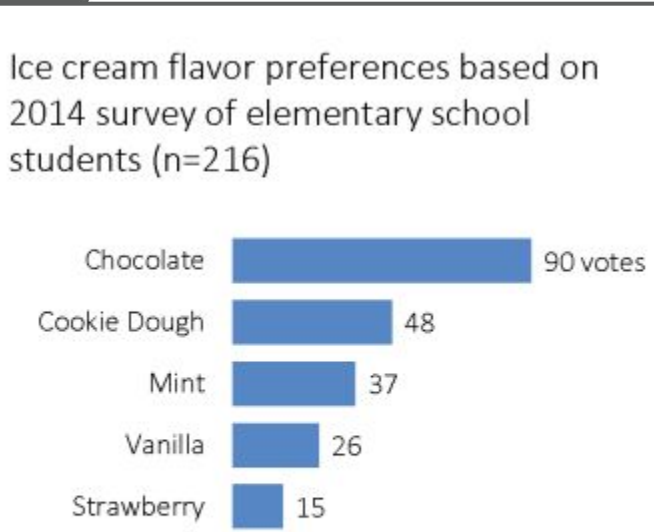
Draw your audience's attention to the most important information.

- Use of colors
- Annotations
- Callouts
- Label critical data points

Use clear titles, axis labels and legends.

Include your takeaways on the slide or in the plot title. Require your audience to have to do as little work as possible to see the takeaway.

Draw Attention to Important Information



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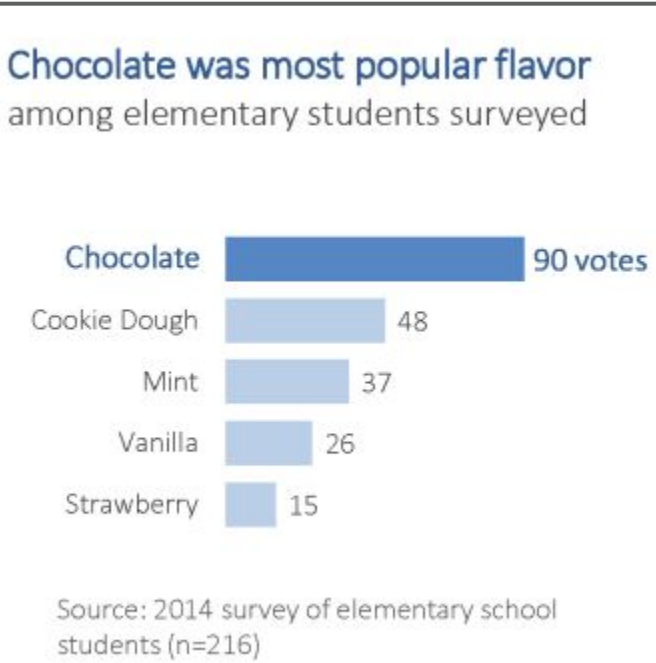

https://exponentphilanthropy.org/blog/show-me-the-story-data-visualization/

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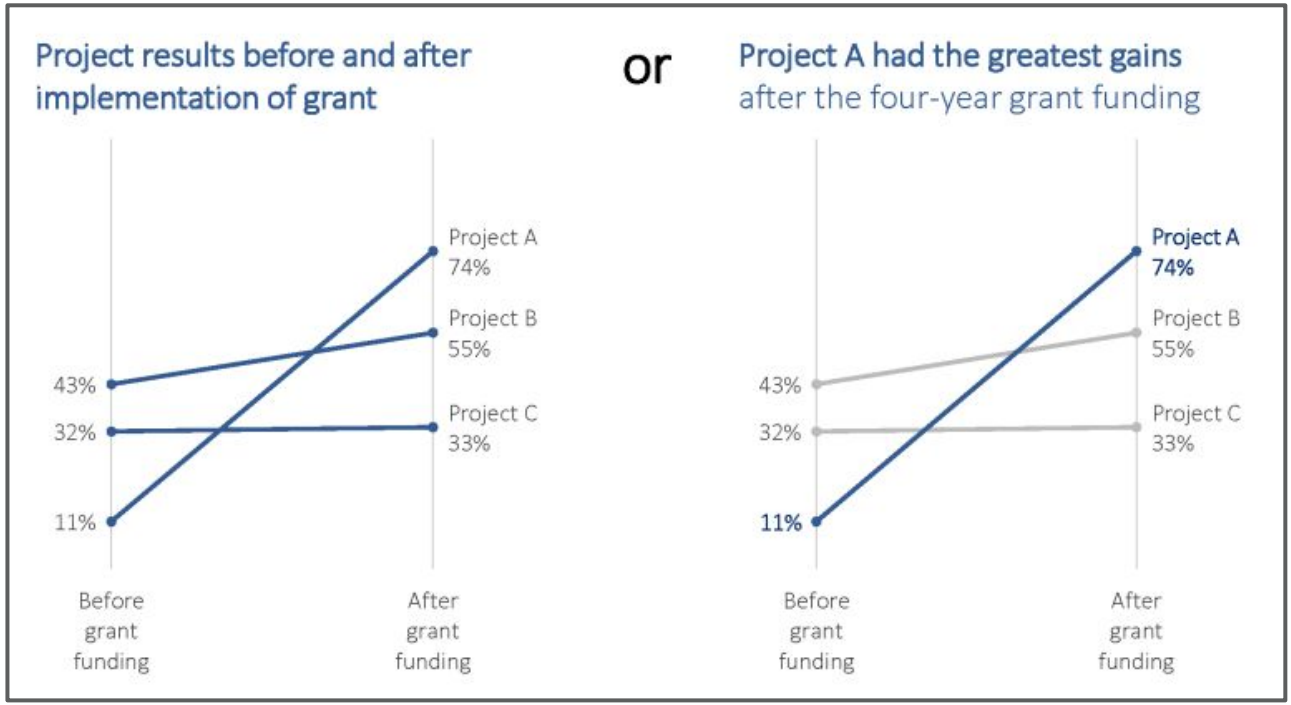
Draw Attention to Important Information



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Presenting a Convincing Argument

Structure your presentation with a clear narrative.

- Frame the context and state the problem or question.
- Sequence insights logically.
- End with key takeaways and, when relevant, practical recommendations.

Use statistics to build your case.

- Reporting confidence intervals or p-values can show that the takeaways not only looks convincing but are statistically sound.

Use confident yet cautious language.

- “Our analysis indicates a likely association.”
- “The data suggests a strong trend.”

Additional Resources

- Python Graph Gallery: https://python-graph-gallery.com/
- Fundamentals of Data Visualization (Book): https://clauswilke.com/dataviz/
- The McCandless Method of Data Presentation: https://artscience.ai/the-mccandless-method-of-data-presentation/
- Data Visualization Guide: https://data.europa.eu/apps/data-visualisation-guide